

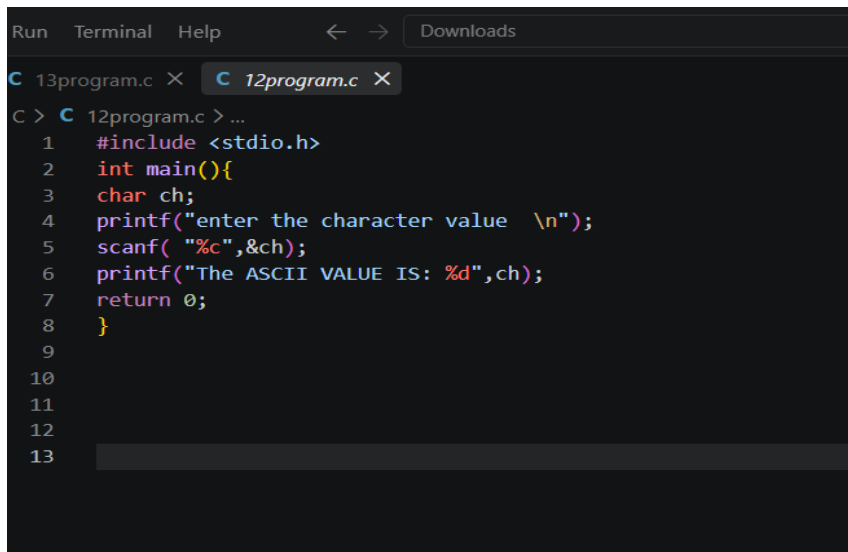
LAB 2

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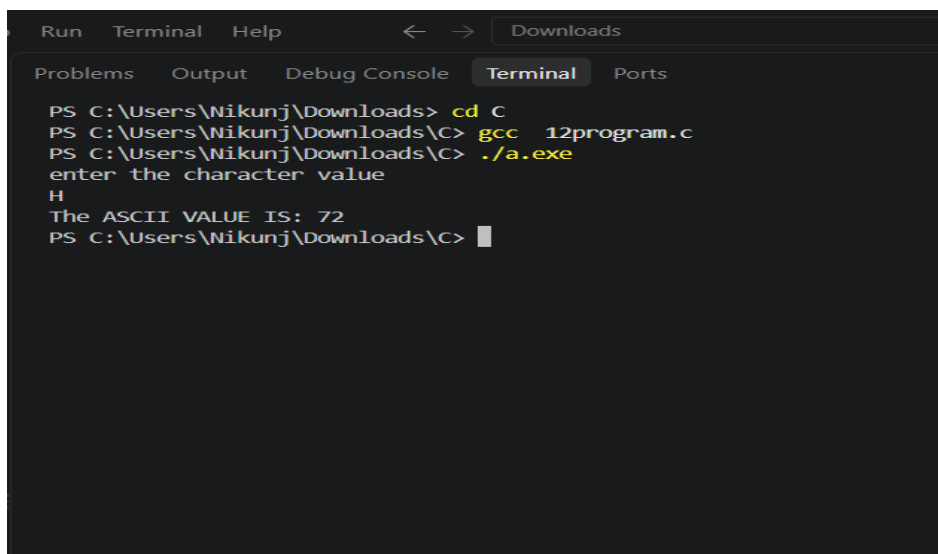
12. PROGRAM TO PRINT ASCII CODE OF ANY CHARACTER.

INPUT:



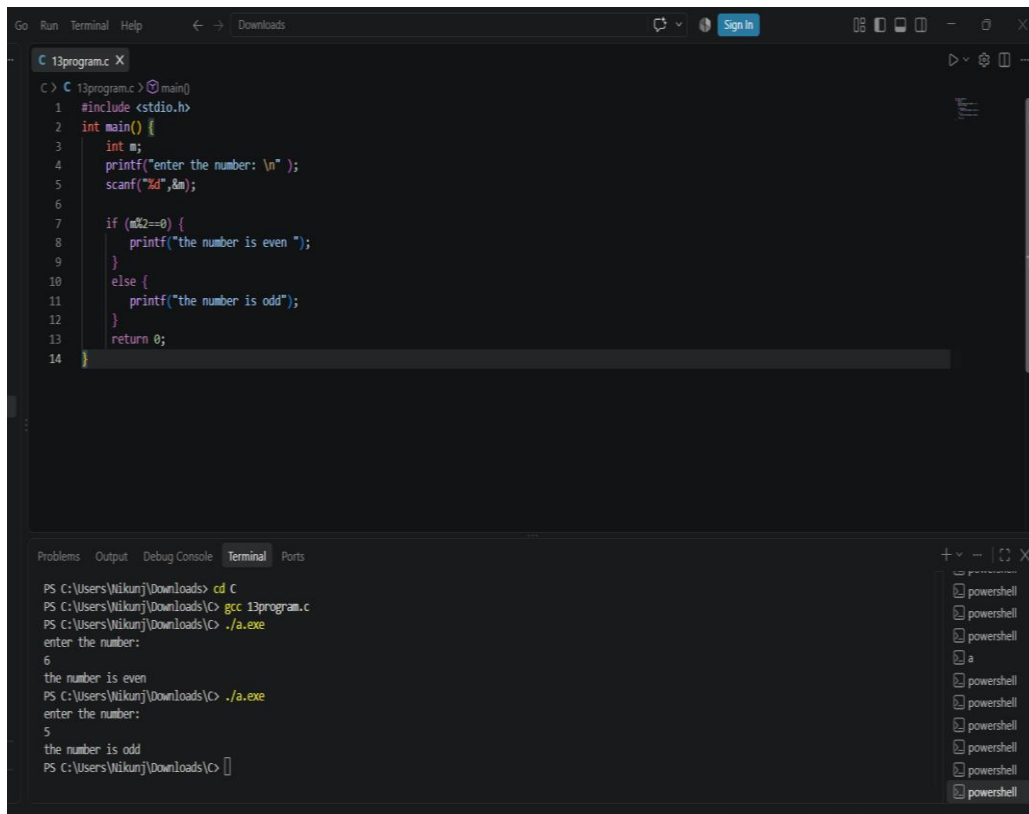
```
Run Terminal Help  Downloads
C 13program.c X  C 12program.c X
C > C 12program.c > ...
1  #include <stdio.h>
2  int main(){
3  char ch;
4  printf("enter the character value \n");
5  scanf( "%c",&ch);
6  printf("The ASCII VALUE IS: %d",ch);
7  return 0;
8  }
9
10
11
12
13
```

OUTPUT



```
Run Terminal Help  Downloads
Problems Output Debug Console Terminal Ports
PS C:\Users\Nikunj\Downloads> cd C
PS C:\Users\Nikunj\Downloads\C> gcc 12program.c
PS C:\Users\Nikunj\Downloads\C> ./a.exe
enter the character value
H
The ASCII VALUE IS: 72
PS C:\Users\Nikunj\Downloads\C> 
```

13. PROGRAM TO CHECK WHETHER A NO. IS EVEN OR ODD.



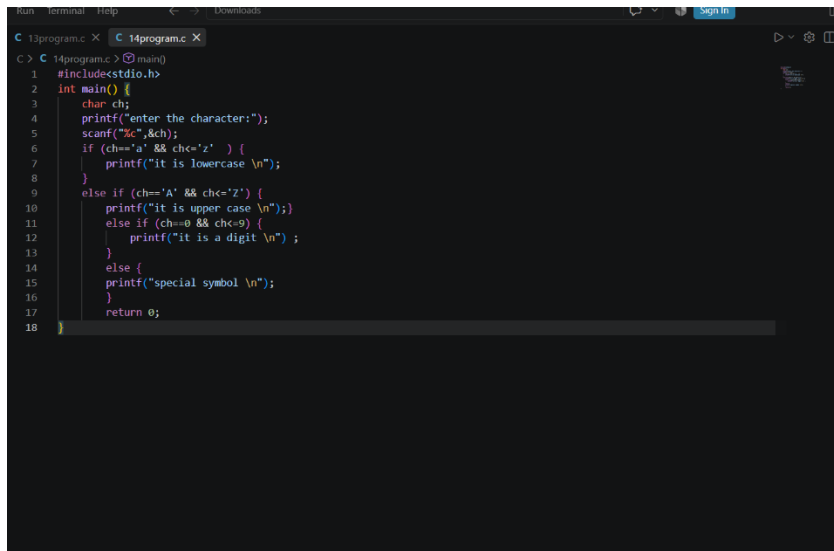
```
C> C: 13program.c X
C> C: 13program.c > (main)
1 #include <stdio.h>
2 int main() {
3     int n;
4     printf("enter the number: \n");
5     scanf("%d",&n);
6
7     if (n%2==0) {
8         printf("the number is even ");
9     }
10    else {
11        printf("the number is odd");
12    }
13    return 0;
14 }
```

Problems Output Debug Console Terminal Ports

```
PS C:\Users\Wilkunj\Downloads> cd C
PS C:\Users\Wilkunj\Downloads> gcc 13program.c
PS C:\Users\Wilkunj\Downloads> ./a.exe
enter the number:
6
the number is even
PS C:\Users\Wilkunj\Downloads> ./a.exe
enter the number:
5
the number is odd
PS C:\Users\Wilkunj\Downloads> 
```

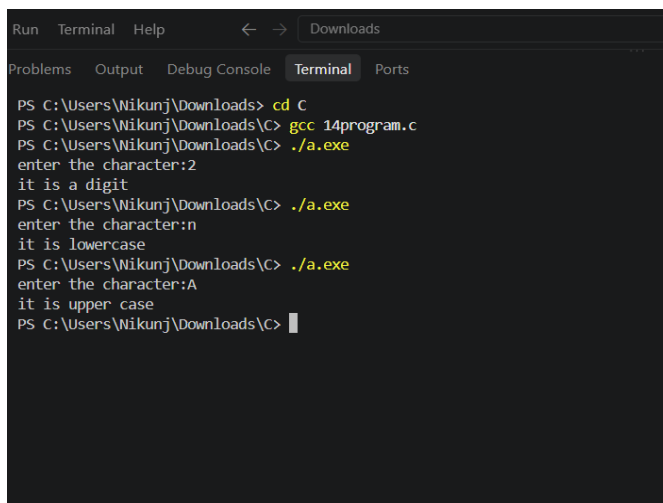
14. PROGRAM TO CHECK WHETHER A GIVEN CHARACTER IS CAPITAL, LETTER, SMALLCASE LETTER, A DIGIT OR A SPECIAL SYMBOL.

INPUT:

A screenshot of a code editor window showing a C program. The program is designed to check the type of a character entered by the user. It includes the standard input/output library and uses a series of conditional statements to determine if the character is lowercase, uppercase, a digit, or a special symbol.

```
C 13program.c X C 14program.c X
C> C 14program.c > @main()
1 #include<stdio.h>
2 int main() {
3     char ch;
4     printf("enter the character:");
5     scanf("%c",&ch);
6     if (ch=='a' && ch<='z' ) {
7         printf("it is lowercase \n");
8     }
9     else if (ch=='A' && ch<='Z') {
10        printf("it is upper case \n");
11    }
12    else if (ch>=0 && ch<=9) {
13        printf("it is a digit \n");
14    }
15    else {
16        printf("special symbol \n");
17    }
18    return 0;
}
```

OUTPUT:

A screenshot of a terminal window showing the execution of the C program. The user enters various characters, and the program outputs the corresponding classification for each. The terminal shows the command prompt, the execution of the compiled program, and the user's input followed by the program's output.

```
Run Terminal Help < -> Downloads
Problems Output Debug Console Terminal Ports
PS C:\Users\Wikunj\Downloads> cd C
PS C:\Users\Wikunj\Downloads\C> gcc 14program.c
PS C:\Users\Wikunj\Downloads\C> ./a.exe
enter the character:2
it is a digit
PS C:\Users\Wikunj\Downloads\C> ./a.exe
enter the character:n
it is lowercase
PS C:\Users\Wikunj\Downloads\C> ./a.exe
enter the character:A
it is upper case
PS C:\Users\Wikunj\Downloads\C> 
```

15 PROGRAM TO READ MARKS OF FIVE SUBJECTS AND PRINT DIVISION.

INPUT:

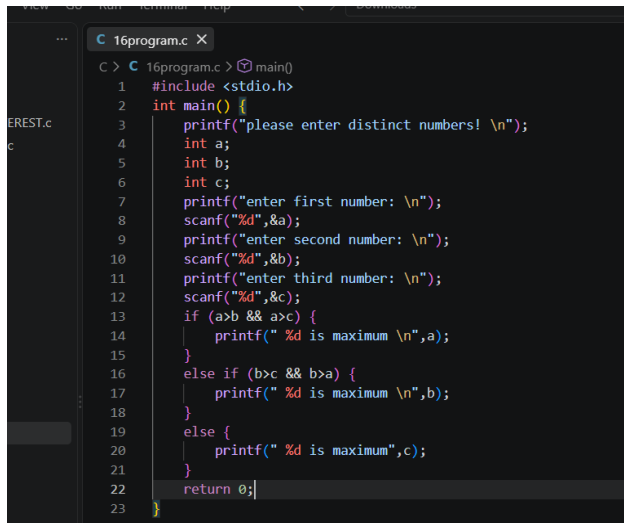
```
C: 15program.c
C> C 15program.c > @ main()
1 #include <stdio.h>
2 int main() {
3     int sub1,sub2,sub3,sub4,sub5,sum,per;
4     printf("enter first subject marks: \n");
5     scanf("%d",&sub1);
6     printf("enter the marks in second subject: \n");
7     scanf("%d",&sub2);
8     printf("enter third subject marks: \n");
9     scanf("%d",&sub3);
10    printf("enter fourth subject marks: \n");
11    scanf("%d",&sub4);
12    printf("enter fifth subject marks: \n");
13    scanf("%d",&sub5);
14    sum=sub1+sub2+sub3+sub4+sub5;
15    per=sum/5;
16    if (per>90) {
17        printf("A+ \n");
18    }
19    else if (per>80 && per<=90) {
20        printf("A \n");
21    }
22    else if (per>70 && per<=80) {
23        printf("B \n");
24    }
25    else if (per>55 && per<=70) {
26        printf("C \n");
27    }
28    else if (per>45 && per<=55) {
29        printf("D \n");
30    }
31    else if (per<=45 && per>=0) {
32        printf("F \n");
33    }
34    else {
35        printf("invalid input");
36    }
37    return 0;
}
```

OUTPUT:

```
Problems Output Debug Console Terminal Ports
PS C:\Users\Witunj\Downloads> cd C
PS C:\Users\Witunj\Downloads\C> gcc 15program.c
PS C:\Users\Witunj\Downloads\C> ./a.exe
enter first subject marks:
55
enter the marks in second subject:
55
enter third subject marks:
22
enter fourth subject marks:
22
enter fifth subject marks:
63
F
PS C:\Users\Witunj\Downloads\C>
```

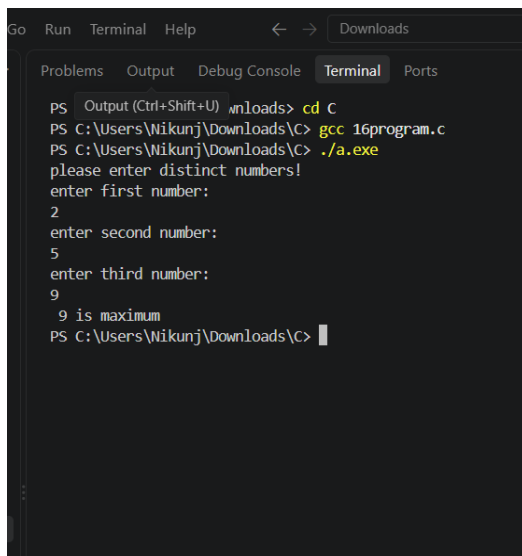
16. PROGRAM TO READ THREE NOS. AND PRINT MAX.

INPUT:

A screenshot of a code editor showing a C program. The editor has a dark theme. The file name is '16program.c'. The code is as follows:

```
C > C 16program.c > (main())
1  #include <stdio.h>
2  int main() {
3      printf("please enter distinct numbers! \n");
4      int a;
5      int b;
6      int c;
7      printf("enter first number: \n");
8      scanf("%d",&a);
9      printf("enter second number: \n");
10     scanf("%d",&b);
11     printf("enter third number: \n");
12     scanf("%d",&c);
13     if (a>b && a>c) {
14         printf(" %d is maximum \n",a);
15     }
16     else if (b>c && b>a) {
17         printf(" %d is maximum \n",b);
18     }
19     else {
20         printf(" %d is maximum",c);
21     }
22     return 0;
23 }
```

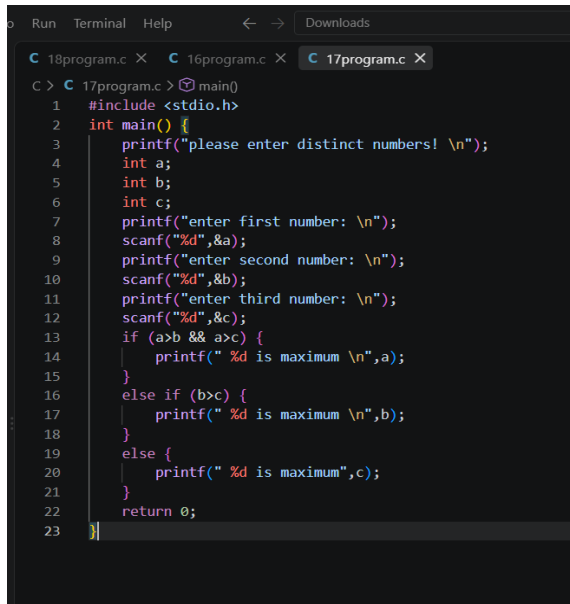
OUTPUT:

A screenshot of a terminal window. The terminal has a dark theme. The command prompt is 'PS C:\Users\Wikunj\Downloads>'. The user has entered 'cd C', 'gcc 16program.c', and './a.exe'. The program output is as follows:

```
PS C:\Users\Wikunj\Downloads> cd C
PS C:\Users\Wikunj\Downloads> gcc 16program.c
PS C:\Users\Wikunj\Downloads> ./a.exe
please enter distinct numbers!
enter first number:
2
enter second number:
5
enter third number:
9
9 is maximum
PS C:\Users\Wikunj\Downloads>
```

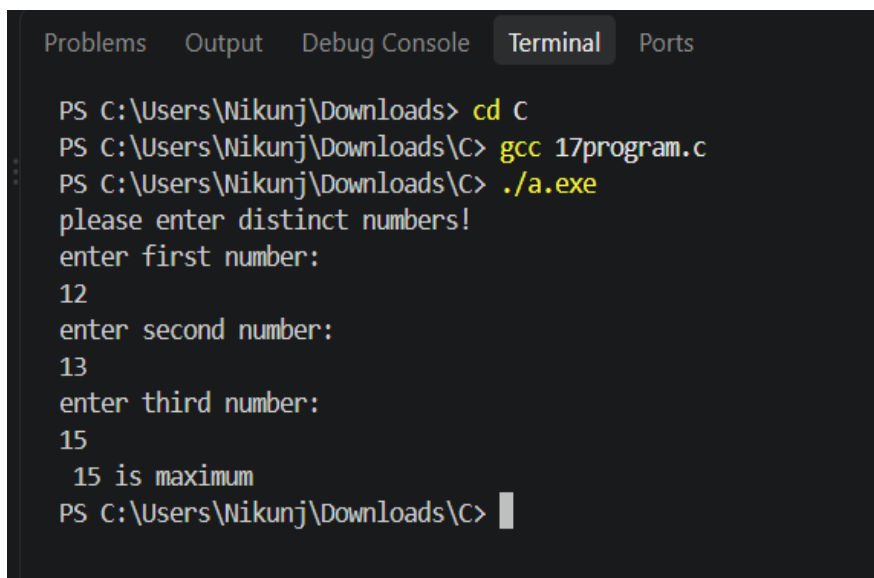
17. PROGRAM TO READ THREE NOS. AND PRINT MAX USING LOGICAL OPERATOR.(&&)

INPUT:

A screenshot of a code editor showing a C program. The editor has tabs for '18program.c', '16program.c', and '17program.c'. The active tab is '17program.c'. The code is as follows:

```
C > C 17program.c > main()
1  #include <stdio.h>
2  int main() {
3      printf("please enter distinct numbers! \n");
4      int a;
5      int b;
6      int c;
7      printf("enter first number: \n");
8      scanf("%d",&a);
9      printf("enter second number: \n");
10     scanf("%d",&b);
11     printf("enter third number: \n");
12     scanf("%d",&c);
13     if (a>b && a>c) {
14         printf(" %d is maximum \n",a);
15     }
16     else if (b>c) {
17         printf(" %d is maximum \n",b);
18     }
19     else {
20         printf(" %d is maximum",c);
21     }
22     return 0;
23 }
```

OUTPUT:

A screenshot of a terminal window with tabs for 'Problems', 'Output', 'Debug Console', 'Terminal', and 'Ports'. The 'Terminal' tab is active. The terminal shows the following commands and output:

```
PS C:\Users\Nikunj\Downloads> cd C
PS C:\Users\Nikunj\Downloads\C> gcc 17program.c
PS C:\Users\Nikunj\Downloads\C> ./a.exe
please enter distinct numbers!
enter first number:
12
enter second number:
13
enter third number:
15
15 is maximum
PS C:\Users\Nikunj\Downloads\C>
```

18. PROGRAM TO READ THREE NOS. AND PRINT MAX USING CONDITIONAL

(?:) OPERATOR.

INPUT:

```
C 18program.c X
C 18program.c > main()
1  #include<stdio.h>
2  int main(){
3  int a,b,c;
4  printf("enter a: \n");
5  scanf("%d",&a);
6  printf("enter b: \n");
7  scanf("%d",&b);
8  printf("enter c: \n");
9  scanf("%d",&c);
10 a>b && a>c ? printf("a= %d is maximum",a):printf("-----");
11 b>c && b>a ? printf("b= %d is maximum",b):printf("-----");
12 c>a && b>a ? printf("c= %d is maximum",c):printf("-----");
13 return 0;
14 }
15
16
```

OUTPUT:

```
PS C:\Users\Nikunj\Downloads\iop> gcc 18program.c
PS C:\Users\Nikunj\Downloads\iop> ./a.exe
enter a:
56
enter b:
65
enter c:
89
-----c= 89 is maximum
PS C:\Users\Nikunj\Downloads\iop> 
```

19. PROGRAM TO READ A CHARACTER AND CHECK WHETHER IT IS A SMALL CASE LETTER OR NOT USING CONDATIONAL OPERATOR.

INPUT:

```
C 19program.c > main()
1  #include <stdio.h>
2  int main() {
3      char ch;
4      printf("enter character : \n");
5      scanf("%c",&ch);
6      ch>='a' && ch<='z'?printf("lower case \n"):printf("not in lower case \n ");
7      return 0;
8
9
10 }
```

OUTPUT:

```
Problems  Output  Debug Console  Terminal  Ports

PS C:\Users\Nikunj\Downloads\iop> gcc 19program.c
PS C:\Users\Nikunj\Downloads\iop> ./a.exe
enter character :
2
not in lower case

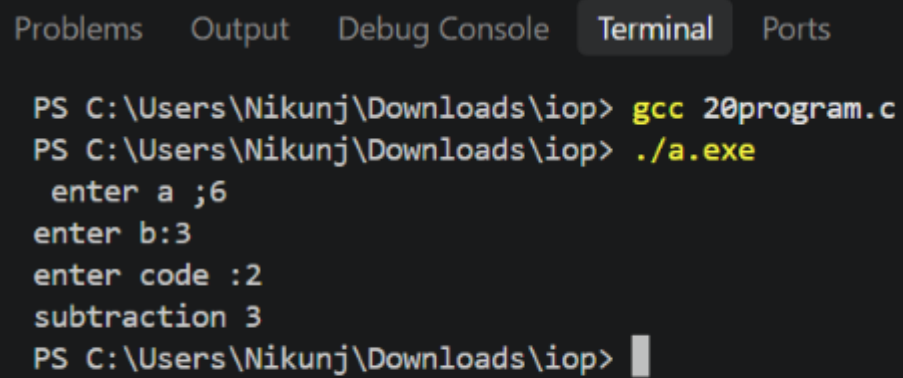
PS C:\Users\Nikunj\Downloads\iop> 
```

20. PROGRAM WHICH WILL WORK LIKE A SIMPLE CALCULATOR USING SWITCH-CASE.

INPUT:

```
C 20program.c X
C 20program.c > ...
1  #include <stdio.h>
2  int main() {
3      int a;
4      int b;
5      int c;
6      printf(" enter a ;");
7      scanf("%d",&a);
8      printf("enter b:");
9      scanf("%d",&b);
10     printf("enter code :");
11     scanf("%d",&c);
12     switch(c) {
13         case 1: printf("addition %d",a+b);
14                 break;
15         case 2: printf("subtraction %d",a-b);
16                 break;
17         case 3: printf("multiplication %d",a*b);
18                 break;
19         case 4: printf("division %d",a/b);
20                 break;
21         default:printf("invalid code");
22     }
23     return 0;
24 }
25
26
27
28 }
```

OUTPUT:



The image shows a screenshot of a terminal window with a dark background. At the top, there is a tab bar with four tabs: 'Problems', 'Output', 'Debug Console', and 'Terminal' (which is selected and highlighted). To the right of the 'Terminal' tab is a 'Ports' tab. The terminal content shows a PowerShell prompt 'PS C:\Users\Nikunj\Downloads\iop>' followed by the command 'gcc 20program.c'. The next line shows the same prompt followed by './a.exe'. The program then prompts the user with 'enter a ;6', 'enter b:3', 'enter code :2', and 'subtraction 3'. The final line shows the prompt 'PS C:\Users\Nikunj\Downloads\iop>' with a white cursor bar.

```
PS C:\Users\Nikunj\Downloads\iop> gcc 20program.c
PS C:\Users\Nikunj\Downloads\iop> ./a.exe
enter a ;6
enter b:3
enter code :2
subtraction 3
PS C:\Users\Nikunj\Downloads\iop> |
```

21. PROGRAM TO READ TWO NOS. AND PERFORM SPECIFIC TASK (USING ARITHMETIC OPERATOR) AND THIS PERFORM USING SWITCH-CASE. (CASE IN CHAR FORM).

INPUT:

```
C 21_program.c X
C 21_program.c > ...
1  #include <stdio.h>
2  int main() {
3      char oper;
4      printf("tell operation to perform :");
5      scanf("%c",&oper);
6      int a;
7      printf("enter first number :");
8      scanf("%d",&a);
9      int b;
10     printf("enter the second number :");
11     scanf("%d",&b);
12
13     switch (oper) {
14         case '+' : printf("%d",a+b);
15             break;
16         case '-' : printf("%d",a-b );
17             break;
18         case '*' :printf("%d",a*b);
19             break;
20         case '/' :printf("%d",a/b);
21             break;
22         default :printf("only + - * / options are available");
23     }
24 }
25
26
27
28
```

OUTPUT:

```
PS C:\Users\Nikunj\Downloads\iop> gcc 21_program.c
PS C:\Users\Nikunj\Downloads\iop> ./a.exe
tell operation to perform :+
enter first number :5
enter the second number :6
11
PS C:\Users\Nikunj\Downloads\iop> 
```

22. PROGRAM TO CALCULATE FACTORIAL OF A NO.

INPUT:

```
C 22program.c X
C 22program.c > ...
1  #include <stdio.h>
2  int main() {
3      int n;
4      printf("enter N:");
5      scanf("%d",&n);
6      int factorial=1;
7      for (int i=n;i>=1;i--){
8          factorial=factorial*i;
9      }
10     printf("factorial of %d is %d",n,factorial);
11     return 0;
12 }
```

OUTPUT:

```
PS C:\Users\Nikunj\Downloads\iop> gcc 22program.c
PS C:\Users\Nikunj\Downloads\iop> ./a.exe
enter N:5
factorial of 5 is 120
PS C:\Users\Nikunj\Downloads\iop> 
```

23. PROGRAM TO READ TWO NOS. AND CALCULATE POWER WITHOUT USING HEADER FILE(<MATH.H>).

INPUT:

```
C 23program.c X
C 23program.c > ...
1  # include <stdio.h>
2  int main(){
3      int base;
4      printf("enter base:");
5      scanf("%d",&base);
6      int power;
7      printf("enter power:");
8      scanf("%d",&power);
9      int ans=1;
10     for (int i=1;i<=power;i++){
11         ans=ans*base;
12     }
13     printf("THE ANSWER IS %d",ans);
14     return 0;
15 }
16
```

OUTPUT:

```
Problems  Output  Debug Console  Terminal  Ports
PS C:\Users\Nikunj\Downloads\iop> gcc 23program.c
PS C:\Users\Nikunj\Downloads\iop> ./a.exe
enter base:5
enter power:6
THE ANSWER IS 15625
PS C:\Users\Nikunj\Downloads\iop>
```